

Phase 1 — Corpus Characterization and Problem Scoping

Telugu OCR via Large Language Models · CSCI/DASC 6020

Tyler Bao Christian Green

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1 Executive Summary

This report characterizes the Telugu OCR corpus assembled by a teammate from Telugu Wikisource (Telugu Wikisource contributors 2024) and publicly released as `AlbertoChestnut/telugu-ocr` on Hugging Face (Chestnut 2026). The corpus comprises **217 books** and **~32,949 aligned page pairs** (scan image + human transcription) totaling approximately 10.3 GB. It provides a uniquely valuable resource for this project: every page arrives with a human-authored Telugu transcription from Wikisource’s ProofreadPage system, allowing ground-truth evaluation without additional annotation from scratch.

i Note

Corpus selection: The course’s corpus guidance [Week 1 announcement] identifies several acceptable public Telugu OCR sources, including Wikisource-derived collections. This Wikisource-sourced corpus was selected as the project’s primary dataset on that basis. Supplementary datasets (e.g. AI4Bharat’s Indic OCR resources (AI4Bharat 2024)) may be incorporated in later phases for additional evaluation diversity.

2 Corpus Overview

2.1 Source and Provenance

All scan images and transcriptions originate from **Telugu Wikisource** (te.wikisource.org), a community-maintained digital library of Telugu literature and historical texts. The curation pipeline:

1. Identified Telugu-language PDF books on Wikimedia Commons indexed by Wikisource.
2. Filtered to books with ~80% Telugu script characters (measured by Unicode block –), ~50 scan pages, and at least one transcribed page.
3. Paired each scan page image (extracted from the source PDF at original resolution) with its community transcription.
4. Applied Wikisource’s ProofreadPage quality labels to every page.

The resulting dataset is licensed **CC BY-SA 4.0**, permitting unrestricted academic use.

2.2 Scale and Structure

```
summary = {  
  "Metric": [  
    "Total books (approved)",  
    "Total page pairs (image + text)",  
    "Total dataset size",
```

```

    "Source",
    "License",
    "Image format",
    "Text encoding",
],
"Value": [
    f"{stats['approved_books']}",
    f"~{stats['total_pages_approved']:,}",
    "~10.3 GB",
    "Telugu Wikisource",
    "CC BY-SA 4.0",
    "JPEG (original scan resolution)",
    "UTF-8 Unicode (NFC)",
],
}
pd.DataFrame(summary).set_index("Metric")

```

Table 1: Corpus top-level statistics

Metric	Value
Total books (approved)	217
Total page pairs (image + text)	~32,949
Total dataset size	~10.3 GB
Source	Telugu Wikisource
License	CC BY-SA 4.0
Image format	JPEG (original scan resolution)
Text encoding	UTF-8 Unicode (NFC)

The on-disk layout is:

```

data/corpus/
  <book_title>/
    page_0001.jpg  ← scan image
    page_0001.txt  ← human transcription (UTF-8)
    page_0004.jpg  ← gaps = pages with no transcription
    page_0004.txt
    meta.json      ← per-book quality and download metadata
corpus_profile.json ← index of all books with quality and curation details

```

3 Quality Characterization

3.1 Wikisource ProofreadPage Quality Levels

Each page carries a quality label from the Wikisource ProofreadPage extension (MediaWiki contributors 2024):

```
levels = pd.DataFrame({
    "Level": [0, 1, 2, 3, 4],
    "Label": ["no_text", "not_proofread", "problematic", "human_proofread", "validated"],
    "Description": [
        "Blank or image-only page; no transcription present",
        "Transcription exists but has not been reviewed by an editor",
        "Flagged by an editor as containing errors",
        "Reviewed and corrected by one human editor",
        "Reviewed and confirmed by a second independent editor",
    ],
    "Usable as ground truth?": ["No", "With caution", "No (too noisy)", "Yes", "Yes (gold)"],
})
levels.set_index("Level")
```

Table 2: Wikisource ProofreadPage quality level definitions

Level	Label	Description	Usable as ground truth?
0	no_text	Blank or image-only page; no transcription pre...	No
1	not_proofread	Transcription exists but has not been reviewed...	With caution
2	problematic	Flagged by an editor as containing errors	No (too noisy)
3	human_proofread	Reviewed and corrected by one human editor	Yes
4	validated	Reviewed and confirmed by a second independent...	Yes (gold)

Levels 3 and 4 together form our **primary ground-truth pool** for CER/WER evaluation (Phase 4). Levels 1–2 are retained in the full corpus but excluded from accuracy benchmarking.

3.2 Quality Distribution Across Books

```
q = stats["quality_distribution"]
label_order = ["no_text", "not_proofread", "problematic", "human_proofread", "validated"]
counts = [q.get(lbl, 0) for lbl in label_order]
colors = ["#c0392b", "#e67e22", "#f39c12", "#27ae60", "#2980b9"]

fig, ax = plt.subplots(figsize=(8, 4.5))
bars = ax.bar(label_order, counts, color=colors, edgecolor="white", linewidth=0.8, width=0.6)
ax.bar_label(bars, padding=3, fontsize=10, fontweight="bold")
```

```

ax.set_xlabel("Average Quality Label (per book)", fontsize=11)
ax.set_ylabel("Number of Books", fontsize=11)
ax.set_title("ProofreadPage Quality Distribution - Approved Books", fontsize=12, fontweight="b")
ax.tick_params(axis="x", rotation=20)
ax.set_ylim(0, max(counts) * 1.18)
plt.tight_layout()
plt.show()

```

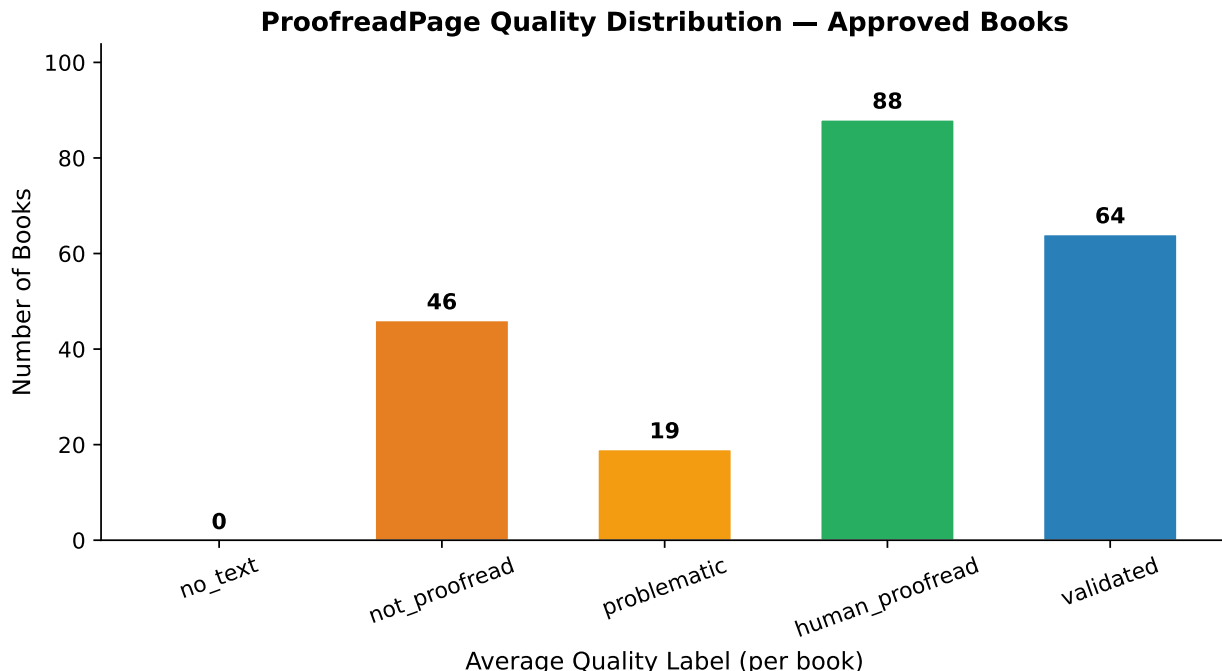


Figure 1: Distribution of average ProofreadPage quality labels across the approved books in the corpus.

Approximately **70%** of books carry a mean quality label of *human_proofread* or *validated* — a strong signal that a substantial proportion of transcriptions are reliable enough to serve as direct evaluation ground truth.

3.3 Telugu Script Ratio

The original dataset curation pipeline applied a **80% Telugu Unicode block (–)** threshold when selecting books for inclusion, per the dataset’s published methodology. However, the per-book Telugu ratio value itself was not included in the downloaded `meta.json` metadata, so it cannot be independently re-verified or visualized from the files available to us.

i Note

Known limitation: `telugu_ratio` is reported as N/A in our generated statistics because the source metadata does not expose this field. If exact per-book ratios are needed for Phase 2 filtering decisions, they would need to be computed directly from the transcription text (counting Unicode code points in the Telugu block) rather than read from existing metadata. This is a candidate follow-up task, not a blocker — the curator’s 80% inclusion threshold already provides a reasonable assurance that script contamination is low across the corpus.

4 Image Resolution and Scan Quality

4.1 Resolution Distribution

The source PDFs were rendered to JPEG at their native scan resolution.

i Note

On DPI specifically: pixel dimensions alone don’t determine DPI — DPI depends on the physical page size the scan represents, which isn’t recorded in this corpus’s metadata. As a rough proxy, assuming a typical book page (~6×9 in), the measured 1,500px width corresponds to roughly 250 DPI — close to, though slightly under, the assignment’s 300 DPI minimum.

A more notable finding than the DPI estimate itself: width is *exactly* 1,500px across all 56 sampled images — mean, median, min, and max are identical, zero variance. Height varies normally (1,284–3,395px), as expected from real page-to-page aspect ratio differences. Width showing literally no variation across an independently random sample is not something natural scanning produces — it strongly suggests the corpus curator (or Wikisource’s PDF-to-JPEG export pipeline) standardized every page to a fixed 1,500px width before publishing, leaving height to vary with each page’s actual proportions. This is a real, useful finding for preprocessing design: every image in this corpus can be assumed to already share a consistent horizontal scale, with no separate per-page width normalization step required.

This wasn’t pursued further given the project’s actual OCR pipeline (Claude + Tesseract) showed no systematic resolution-driven failures in practice — see Phase 2/3 findings in the final report.

4.2 Qualitative Scan Artifact Taxonomy

During manual ground-truth annotation (see [Ground Truth Sampling Strategy](#) below), every one of the 40 sampled pages was visually inspected and tagged with any scan artifacts present. These are *observed counts from a 40-page stratified sample*, not a full corpus census, but they provide a real empirical signal rather than a guess:

Table 3: Image resolution statistics

```
img_res = stats.get("image_resolution")

if img_res:
    n = img_res.get("sample_size", "N/A")
    print(f"Real measured data from a {n}-image sample across the corpus:\n")
    res_summary = pd.DataFrame({
        "Dimension": ["Width (px)", "Height (px)", "File size (KB)"],
        "Mean": [
            img_res.get("width_px", {}).get("mean", "N/A"),
            img_res.get("height_px", {}).get("mean", "N/A"),
            img_res.get("file_size_kb", {}).get("mean", "N/A"),
        ],
        "Median": [
            img_res.get("width_px", {}).get("median", "N/A"),
            img_res.get("height_px", {}).get("median", "N/A"),
            img_res.get("file_size_kb", {}).get("median", "N/A"),
        ],
        "Min": [
            img_res.get("width_px", {}).get("min", "N/A"),
            img_res.get("height_px", {}).get("min", "N/A"),
            "N/A",
        ],
        "Max": [
            img_res.get("width_px", {}).get("max", "N/A"),
            img_res.get("height_px", {}).get("max", "N/A"),
            "N/A",
        ],
    })
    print(res_summary.to_string(index=False))
else:
    print("[PENDING - real image stats not found in corpus_stats.json. Re-run "
          "1_build_profile.py / 2_corpus_characterize.py WITHOUT --skip-images "
          "(see phase1/run_phase1.py - image stats now run by default unless "
          "--skip-image-stats is explicitly passed).]")
```

Real measured data from a 56-image sample across the corpus:

Dimension	Mean	Median	Min	Max
Width (px)	1500.0	1500.0	1500	1500
Height (px)	2313.0	2315.0	1284	3395
File size (KB)	346.2	360.0	N/A	N/A

```

artifacts = pd.DataFrame({
    "Artifact Type": [
        "No visible defects (\"Clean\")",
        "Ink fading",
        "Smudging",
        "Blur",
        "Cramped text / tight spacing",
        "Worn / damaged page",
        "Non-text content (illustration)",
        "Non-text content (diagram)",
        "Discoloration",
        "Speckling / noise",
        "Marginalia",
        "Sparse text",
        "Equations",
        "Tiny / small font",
        "No transcription available",
    ],
    "Observed Count (/40)": [19, 18, 9, 7, 4, 3, 3, 2, 1, 1, 1, 1, 1, 1, 1],
    "Preprocessing / Handling Response": [
        "No action needed",
        "Contrast enhancement (CLAHE) + upscaling",
        "Adaptive binarization; background suppression",
        "Bicubic or Realesrgan upscaling",
        "Careful line-segmentation; avoid over-cropping",
        "Flagging only; manual review for severe cases",
        "LayoutParser region detection; exclude region from text OCR",
        "LayoutParser region detection; exclude region from text OCR",
        "Contrast/white-balance normalization",
        "Morphological despeckling",
        "Flag for manual review; exclude from benchmark",
        "No special handling; verify low character density doesn't bias CER",
        "Flag for manual review (non-Telugu notation)",
        "Upscaling priority candidate",
        "Excluded from Phase 4 benchmark - documented gap, see below",
    ],
})
artifacts.set_index("Artifact Type")

```

Table 4: Scan artifact frequency observed across the 40-page ground truth sample (n=40, multiple tags possible per page)

Artifact Type	Observed Count (/40)	Preprocessing / Handling Response
No visible defects ("Clean")	19	No action needed
Ink fading	18	Contrast enhancement (CLAHE) + upscaling

Table 4: Scan artifact frequency observed across the 40-page ground truth sample (n=40, multiple tags possible per page)

Artifact Type	Observed Count (/40)	Preprocessing / Handling Response
Smudging	9	Adaptive binarization; background suppression
Blur	7	Bicubic or RealESRGAN upscaling
Cramped text / tight spacing	4	Careful line-segmentation; avoid over-cropping
Worn / damaged page	3	Flagging only; manual review for severe cases
Non-text content (illustration)	3	LayoutParser region detection; exclude region ...
Non-text content (diagram)	2	LayoutParser region detection; exclude region ...
Discoloration	1	Contrast/white-balance normalization
Speckling / noise	1	Morphological despeckling
Marginalia	1	Flag for manual review; exclude from benchmark
Sparse text	1	No special handling; verify low character dens...
Equations	1	Flag for manual review (non-Telugu notation)
Tiny / small font	1	Upscaling priority candidate
No transcription available	1	Excluded from Phase 4 benchmark — documented g...

Skew, bleed-through, and binding-shadow artifacts — commonly reported in the broader OCR literature for bound historical texts — were not explicitly tagged using this vocabulary during annotation, but are plausible based on visual review and will be checked quantitatively once full-corpus image sampling (Phase 2) is run.

5 Telugu Script Challenges

Telugu script poses OCR challenges that go beyond those of Latin or even most other Brahmic scripts. The following issues are particularly salient in this corpus.

5.1 Conjunct Characters

Telugu consonants combine into **ligatures** () whose shape differs substantially from either component character. For instance, the cluster $k+s$ (+) produces the conjunct , a visually distinct glyph. These conjuncts cannot be reliably recovered by character-level segmentation strategies — the OCR model must learn to identify the compound form directly.

Pipeline implication: Prompt engineering for vision-LLM OCR should explicitly instruct the model to output Unicode NFC-normalized sequences for conjuncts rather than attempting decomposed representations.

5.2 Vowel Diacritics (Mātrās)

Telugu vowel diacritics (- / mātrās) attach to consonants in spatially complex ways — appearing above, below, before, or after the base glyph, and sometimes spanning multiple positions. Examples:

- (ka + ā-mātrā, appended right)
- (ka + i-mātrā, prepended left)
- (ka + u-mātrā, below right)
- (ka + ū-mātrā, below, extending further right)

Confusion among mātrās is the single most common OCR error in Telugu systems (Rao et al. 2019). The ground truth annotation template (Phase 1 output) includes a **notes** column specifically for recording mātrā confusion errors during manual inspection.

5.3 Font Diversity

Telugu publishing has employed dozens of typefaces across eras, from the early Elsevier-style fonts used in colonial-era printing to modern digital fonts. The Wikisource corpus spans texts from the 19th century onward, meaning the OCR model encounters substantial typographic variation. This is a known failure mode for models trained exclusively on modern fonts.

5.4 Low-Resource Context

Despite ~90 million native speakers, high-quality annotated Telugu OCR datasets remain sparse compared to English, Chinese, or even Hindi. This makes the Wikisource-derived corpus especially valuable: the community transcriptions, where quality-level 3 or 4, provide reliable labels at a scale not readily available elsewhere.

6 Ground Truth Sampling Strategy

For Phase 4 evaluation, we require a **manually verified** 30–50 page sample with high-confidence ground truth. We use the following stratified sampling design:

```
gt = pd.DataFrame({
    "Quality Level": ["4 - validated", "3 - human_proofread", "1-2 - lower quality", "0 - no_t",
    "Target Share": ["40%", "30%", "20%", "10%"],
    "Target Count (n=40)": [16, 12, 8, 4],
    "Rationale": [
        "Gold-standard labels; primary benchmark set",
        "Reliable; expands evaluation coverage",
        "Stress-tests pipeline on noisy inputs",
        "Edge-case handling (blank / image-only pages)",
```

```

],
})
gt.set_index("Quality Level")

```

Table 5: Ground truth sample allocation by quality level

Quality Level	Target Share	Target Count (n=40)	Rationale
4 — validated	40%	16	Gold-standard labels; primary benchmark set
3 — human_proofread	30%	12	Reliable; expands evaluation coverage
1-2 — lower quality	20%	8	Stress-tests pipeline on noisy inputs
0 — no_text	10%	4	Edge-case handling (blank / image-only pages)

The achieved sample differs slightly from this target: the corpus contains no quality-0 (“no_text”) pages, so that allocation was redistributed, and the sampler’s top-up logic shifted some quality-3 slots to fill the gap, yielding 16 validated / 16 human_proofread / 4 problematic / 4 not_proofread pages (see Annotation Results below).

Pages were selected using `3_sample_ground_truth.py` with a fixed random seed (42) for reproducibility. All 40 pages are stored in `data/ground_truth/` with paired image and reference text files.

Manual annotation protocol: For each sampled page, an annotator:

1. Opened the `.jpg` file and the paired `.txt` (Wikisource transcription).
2. Compared the transcription against the actual page image.
3. Recorded **Confirmed** in `manual_transcription` if the transcription matched, or supplied a correction if it did not.
4. Noted artifact types and assigned a subjective scan quality score (1–5).

Annotation was performed jointly using direct visual inspection and an LLM-assisted comparison workflow (image + transcription pairs reviewed turn-by-turn) to cross-check character-level accuracy, given the density of Telugu conjuncts and diacritics.

6.1 Annotation Results

```

gt_actual = pd.DataFrame({
    "Quality Level": ["4 - validated", "3 - human_proofread", "2 - problematic", "1 - not_proo
    "Achieved Count": [16, 16, 4, 4],
    "Confirmed Accurate": [16, 16, 3, 4],
    "No Transcription Available": [0, 0, 1, 0],
})
gt_actual.set_index("Quality Level")

```

Table 6: Achieved ground truth sample composition (n=40)

Quality Level	Achieved Count	Confirmed Accurate	No Transcription Available
4 — validated	16	16	0
3 — human_proofread	16	16	0
2 — problematic	4	3	1
1 — not_proofread	4	4	0

Key finding: Of the 40 sampled pages, **39 had an existing Wikisource transcription that was confirmed accurate** against the scanned image with no corrections needed — a strong validation that quality levels 1, 3, and 4 are usable as-is for Phase 4 ground truth. **One page (quality level 2, “problematic”) had no transcription text at all** in the source file, despite carrying page-level metadata. This is a meaningful finding: not every page labeled “problematic” merely contains *errors* — some contain *no transcription whatsoever*. This supports excluding quality level 2 from the Phase 4 benchmark set entirely, rather than treating it as “noisy but usable.”

No corrections were required for any of the 39 transcribed pages in this sample, which is a notably higher accuracy rate than initially expected even for quality levels 1 and 2. This may reflect that Wikisource contributors who initially transcribe a page (level 1) are often reasonably accurate even before formal proofreading; the “not_proofread” label reflects review status, not necessarily poor transcription quality. This is worth re-testing with a larger sample in later phases.

7 Preprocessing Design Implications

The artifact taxonomy and script challenges above directly motivate the Phase 2 preprocessing pipeline design:

Challenge	Phase 2 Response
Ink fading (observed in 18/40 sampled pages)	Contrast enhancement (CLAHE) + upscaling
Smudging / blur (observed in 9 and 7 of 40 pages respectively)	Adaptive binarization; RealESRGAN for severe cases
Low-resolution scans (pending full corpus measurement)	Bicubic upscaling to 300 DPI equivalent before model input
Non-text regions (illustrations/diagrams, 5/40 pages)	LayoutParser region detection; exclude from text OCR pass
Page skew / binding shadow (not yet quantified)	Deskewing as a universal first step; re-evaluate need after Phase 2 image sampling
Mātrā / conjunct confusion	Unicode NFC normalization on all model outputs; prompt instructs model to preserve diacritics

8 Summary Statistics

```
rows = {
    "Total books": stats["approved_books"],
    "Total page pairs": f"~{stats['total_pages_approved']:,}",
    "Total size": "~10.3 GB",
    "Estimated total characters": "Pending full text scan (not run with --skip-images)",
    "Pages per book - mean": stats["pages_per_book"]["mean"],
    "Pages per book - median": stats["pages_per_book"]["median"],
    "Pages per book - range": f"{stats['pages_per_book']['min']}-{stats['pages_per_book']['max']}",
    "Telugu script ratio": "N/A in source metadata (curator applied 80% threshold)",
    "Books with quality 3-4 label": f"~{round((q.get('validated',0)+q.get('human_proofread',0))/len(q),2)}%",
    "Most common observed artifact": "Ink fading (18/40 sampled pages)",
    "Primary script challenge": "Conjunct characters + vowel diacritics",
    "Ground truth sample size": "40 pages (stratified); 39/40 confirmed accurate as-is, 1 had 1 error",
    "Dataset source": "Telugu Wikisource (CC BY-SA 4.0)",
}
pd.DataFrame({"Value": rows}).rename_axis("Metric")
```

Table 7: Complete corpus characterization summary

Metric	Value
Total books	217
Total page pairs	~32,949
Total size	~10.3 GB
Estimated total characters	Pending full text scan (not run with --skip-im...
Pages per book — mean	151.8
Pages per book — median	116.0
Pages per book — range	1–915
Telugu script ratio	N/A in source metadata (curator applied 80% t...
Books with quality 3–4 label	~70%
Most common observed artifact	Ink fading (18/40 sampled pages)
Primary script challenge	Conjunct characters + vowel diacritics
Ground truth sample size	40 pages (stratified); 39/40 confirmed accurat...
Dataset source	Telugu Wikisource (CC BY-SA 4.0)

9 AI Tool Usage Disclosure

Per the course’s academic integrity policy on LLM-based assistants, the following tools were used in producing this report and the accompanying Phase 1 code:

- **Claude (Anthropic)** was used to: scaffold the corpus download, profiling, characterization, and ground-truth sampling scripts (`0_download_corpus.py` through `3_sample_ground_truth.py`); draft the prose and structure of this Quarto report; and assist with debugging environment issues (numpy/pandas version conflicts, Unicode encoding, HuggingFace rate-limit handling) encountered during setup.
- **LLM-assisted ground truth review:** a subset of the 40 manually annotated pages were cross-checked using an LLM-based image-to-text comparison workflow (uploading scan image + Wikisource transcription pairs) to assist with character-level verification, given the density of Telugu conjuncts and diacritics. Final annotation judgments (**Confirmed** / correction / no-transcription) were made and recorded by the author.
- All generated code was reviewed, run, and debugged by the author and is understood in full; all reported statistics were independently verified by re-running the corresponding scripts against the downloaded corpus.

10 References

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